

Financial Variance & Diagnostic Report

ENTITY: Project-Vessel-2026

PERIOD: 2025-Q2

CURRENCY: NOK

TYPE: Automated Analysis & Process Blueprint

Q2 2025 Executive Summary

**Revenue:
Down (-10.0%)**



**Costs:
Up (+10.5%)**



Core Drivers

- Customer QA milestone verification delayed, deferring billing to Q3.
- Propulsion subcontractor enforced material price indexation.

Variance Diagnostic Matrix

	Actual	Budget	Variance	
Revenue:	45,000,000	50,000,000	-5,000,000 (-10.0%)	EV-DOC-MILESTONE
Cost:	42,000,000	38,000,000	+4,000,000 (+10.5%)	EV-DOC-PROPULSION

Currency: NOK throughout.

Retrieved Root Cause Evidence

DOC-MILESTONE-REV-02

Ultrasonic testing
milestone certificate
postponement delayed
revenue recognition
into Q3.

DOC-PROPULSION-01

Propulsion waterjet
and turbine integration
cost overrun due to
subcontractor **price
indexation**.

Prescribed Investigations

[Action Type: Investigate]

Revenue Recovery

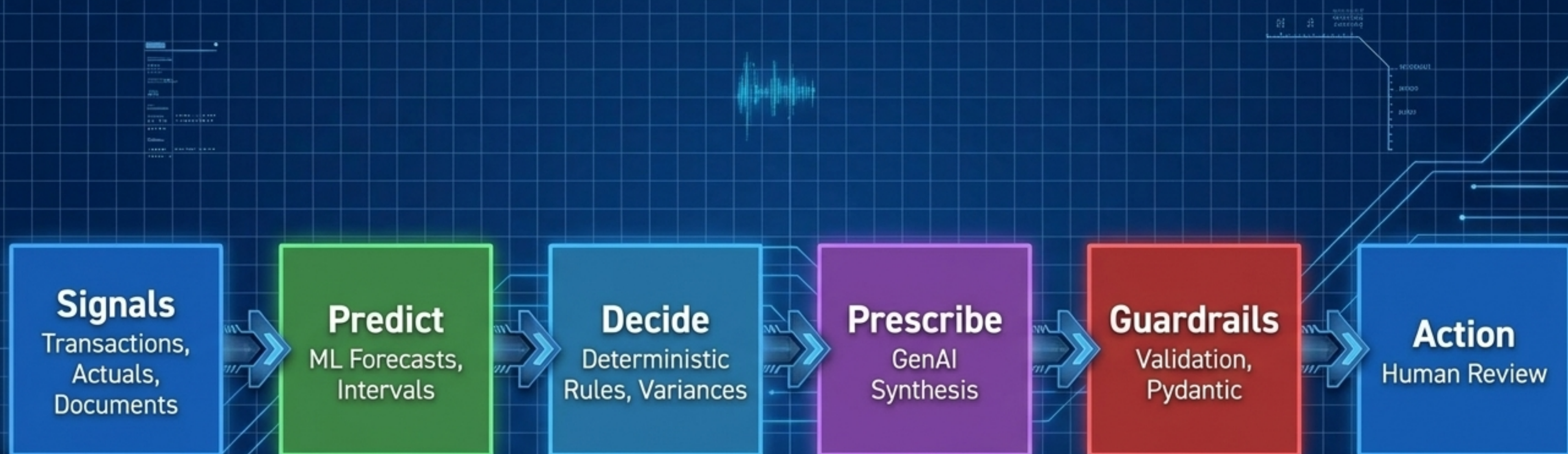
Reconcile customer QA certificate acceptance for the deferred Q2 milestone billing in Q3.

[Action Type: Investigate]

Cost Control

Review subcontractor contractual indexation caps under clause 14.3.

The Predict-to-Prescribe Architecture



The Boundary: Authoritative vs. Generative

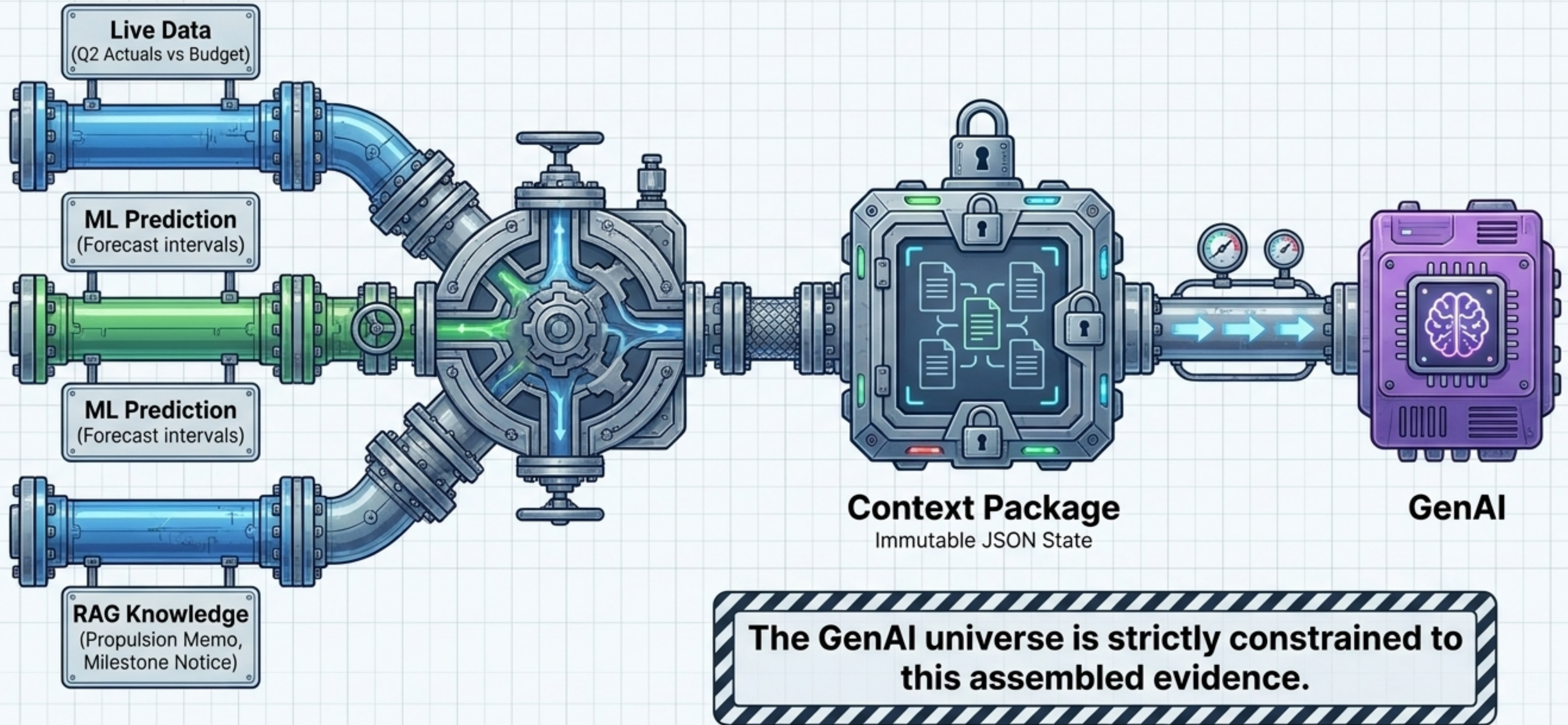
ALLOWED - CrewAI Role

- ✓ Orchestration & Sequencing
- ✓ Context Interpretation
- ✓ Evidence Synthesis & Reporting

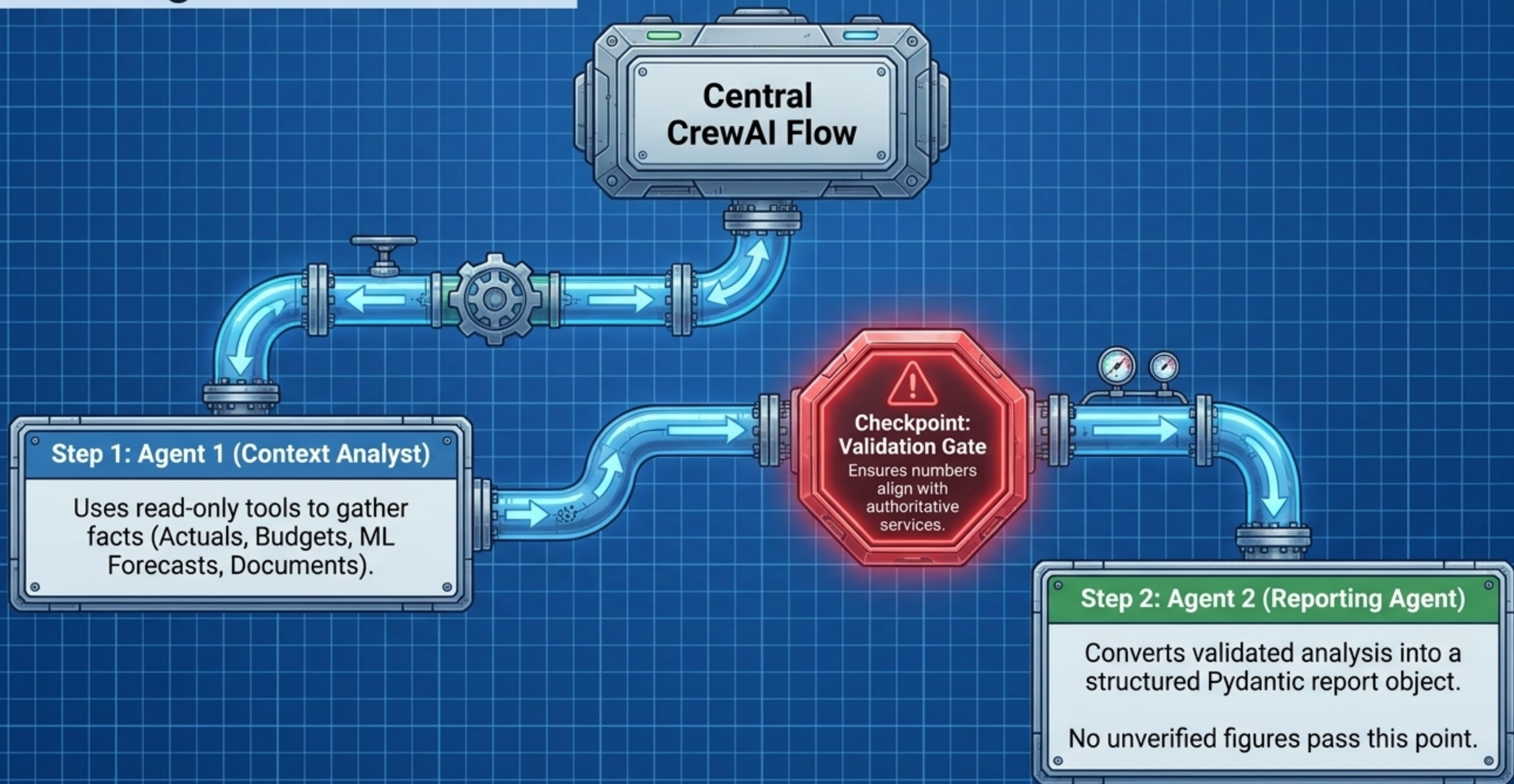
FORBIDDEN - Strict Boundaries

- ✗ Executing arbitrary SQL / Python
- ✗ Inventing or altering numeric forecasts
- ✗ Executing autonomous financial transactions

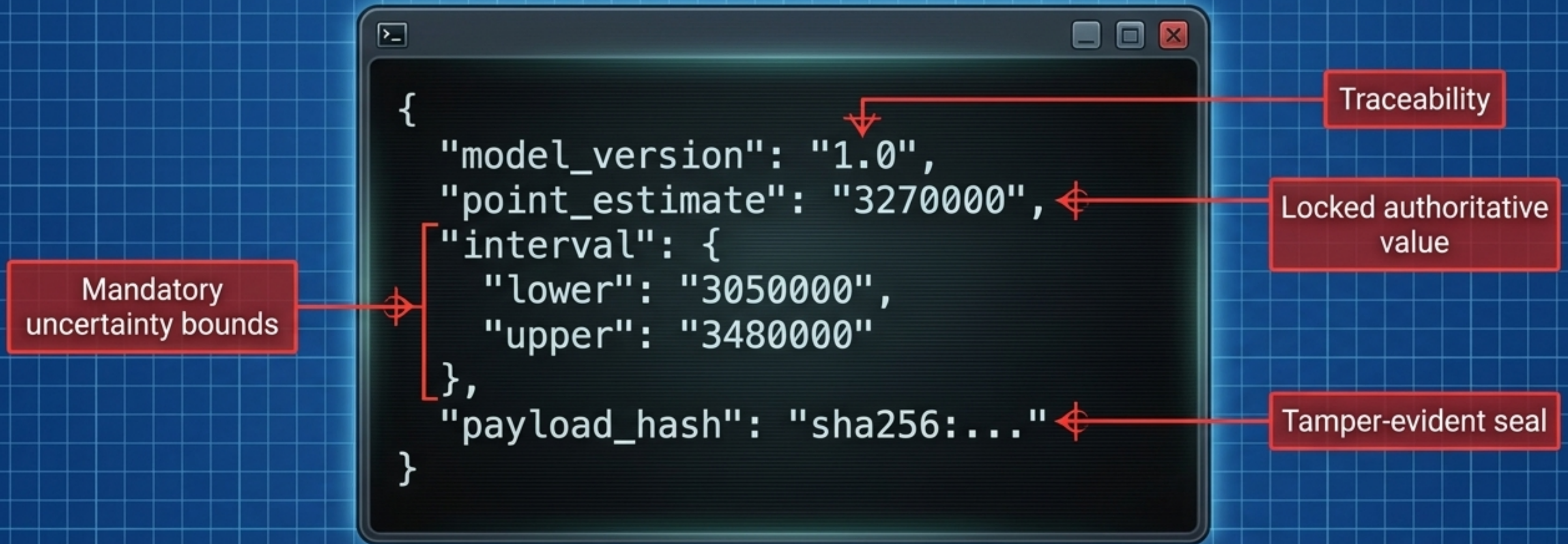
Injecting Reality Before Generation



Multi-Agent Orchestration

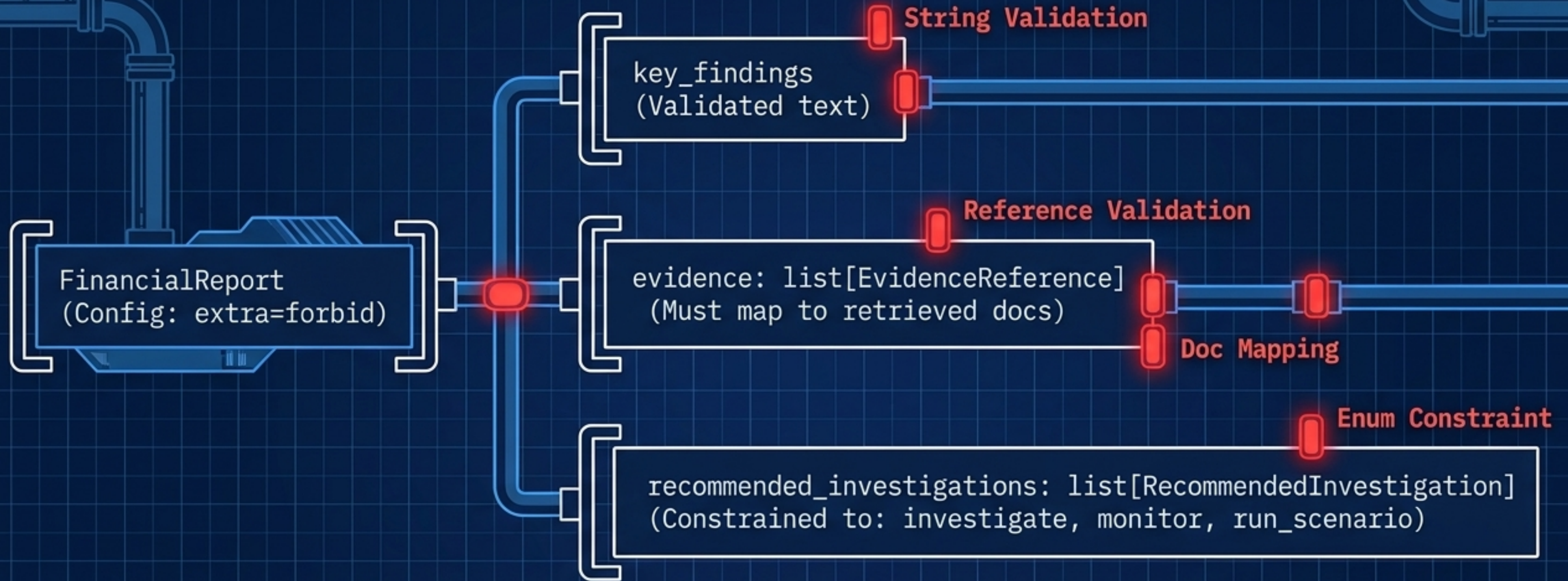


The Immutable Prediction Contract



Key Insight: The LLM receives a read-only serialization. It may explain it, but it cannot recalculate or overwrite prediction fields.

Strict Output Models (Pydantic)



Key Insight: No free-form hallucinations allowed. If the LLM generates an unapproved action type, the system rejects it deterministically.

Closing the Learning Loop



System Trust is Built Through Boundaries

Deterministic Authority

Financial math, records, and predictions remain strictly in code.

Context, Not Calculation

GenAI is utilized exclusively for reasoning, evidence synthesis, and natural language explanation.

Continuous Validation

Human-in-the-loop review, strict output contracts, and continuous offline evaluation guard against system drift and excessive agency.